2017 EXAMINATIONS

Lancaster University

Part II

COMPUTING AND COMMUNICATIONS [2.5 Hours]

SCC.365 Advanced Networking

Candidates are asked to answer THREE questions from FOUR; each question is worth a total of 25 marks.

Use a separate answer book for each question.

Question 1

1.a

i. The Lancaster University Campus Network is a complex configuration of switches and routers, providing connectivity to many thousands of devices and end users. Describe the advantages and disadvantages of introducing a Quality of Service architecture such as Differentiated Services into the campus network.

[4 marks]

ii. Beyond the campus, networks such as JANET and GEANT scale upwards to support many millions of users across the UK and Europe. Describe the advantages and disadvantages of using Integrated Services and RSVP in such an environment.

[4 marks]

1.b The area of QoS management can be broadly divided into *static* and *dynamic* QoS management.

i. Briefly describe the static QoS management techniques of *mapping*, *negotiation*, *admission control* and *resource allocation*.

[4 marks]

ii. Describe three *dynamic* QoS management techniques.

[6 marks]

1.c Considering the Differentiated Services model of QoS support:

i. Describe the difference between intra-domain and inter-domain resource management.

[2 marks]

ii. Describe the main characteristics of the Expedited Forwarding Per-Hop Behaviour (EF PHB).

[3 marks]

iii. What applications are suited to utilising a Less-than Best Effort (LBE) service?

[2 marks]

Total 25 marks

Question 2

2.a At the highest level of abstraction, the architecture of a router is divided into 4 key components. Describe the main activities of each of the components.

[8 marks]

2.b A "QoS capable router" describes a router that is able to control both loss differentiation and delay differentiation, to support a particular set of service levels. Using a diagram, outline the main traffic control mechanisms within such a router, and describe the function of each of the mechanisms.

[10 marks]

2.c A token bucket algorithm can be used to check that packets in a network conform to defined bandwidth and burstiness limits. In terms of basic principles, outline how a token bucket algorithm works.

[3 marks]

2.d A layer-N flow is a sequence of packets with one common layer-N characteristic. Use an example to describe how a flow can be identified at layer 3. Repeat the task for a flow at layer 4.

[4 marks]

Total 25 marks

Question 3

Software Defined Networking (SDN) and Networking Functions Virtualisation (NFV) are both relatively new concepts within the networking industry.

3.a Describe four key requirements of a modern network, and explain why *you* believe these particular requirements are necessary.

[4 marks]

3.b You have been commissioned by a company to investigate the use of SDN and OpenFlow within their business. More specifically, you have been asked to recommend whether the company should introduce SDN using *Pure OpenFlow* or *Hybrid OpenFlow* approaches. Explain, in detail, the principles of each approach, the steps required for the business to introduce SDN in each case, and your overall recommendation.

[7 marks]

3.c OpenFlow can operate in either *reactive* or *proactive* mode. Explain the behaviour of each mode, and the associated trade-offs between each approach.

[4 marks]

3.d You have been commissioned by an ISP to write a report outlining why they would benefit from using NFV. Explain the core concepts of NFV and describe where and how it could leave to potential savings within their business.

[6 marks]

3.e Intent Driven Networking is an area of networking research that is currently receiving significant attention. Explain the concepts of Intent Driven Networking, and provide a worked scenario where the use of this approach would be beneficial.

[4 marks]

Total 25 marks

Question 4

4.a It is predicted that "Content Delivery Networks (CDNs) will carry nearly two-thirds of all internet video traffic by 2020". Suggest why a content provider would choose to use a CDN.

[1 mark]

4.b Using a diagram, explain in detail how CDNs operate in the general case of a user requesting a video from a company that uses a CDN provider to host their content.

[4 marks]

4.c Netflix and Google's YouTube are both examples of video streaming services that make use of Content Distribution Networks (CDNs). Describe two key differences between Netflix and Google's YouTube in the approach they use to content distribution.

[2 marks]

4.d Dynamic Adaptive Streaming over HTTP (DASH) is a relatively new standard enabling the high-quality streaming of video content across the Internet. Imagine a scenario where a DASH client is receiving a high-quality video stream, but the network conditions temporarily change such that the client is no longer able to continue receiving this quality level. Using your knowledge of DASH, explain how the client would behave under these conditions.

[4 marks]

4.e Explain asymmetric routing and how it may arise in the case of a small consumer broadband access ISP. How might asymmetric routing impact on an ISPs network design and peering strategy?

[4 marks]

4.f Imagine you are a network designer at a *large national broadcast company*.

i. In choosing how to route your user traffic what two types of factor are you most concerned about?

[2 marks]

ii. Based on these two factors, discuss the merits of connecting and routing your subscriber traffic directly to:

a) national ISPs,
b) an internet exchange point,
c) content delivery networks (CDNs), or
d) tier 1 transit providers?

[8 marks]

Total 25 marks

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